

DATA MINING AND WAREHOUSING

Introduction

From the time immemorial humans have been manually extracting hidden predictive patterns from data, but the increasing volume of data in modern time requires an automatic approach. With the advent of data mining, it provides a new powerful technology with great potential to help private and public focus on the most important information in their data bases. Data mining is a result of a long process of research and product development, and the primary reason is to assist not only in uncovering hidden patterns from databases but also consists of collecting, managing, analysis and prediction of data.

The term data mining derived its name from the similarities between searching for valuable information in a large database and mining a mountain for a vein of valuable one so the two processes require either sifting through an immense amount of material, or intelligently probing it to find where the value resides. This unit examines the meaning of data mining, the difference between it and knowledge discovery in databases (KDD), evolution of data mining, its scope, architecture and how it works.

Definition of Data Mining

Data mining is an analytical process designed for extracting or exploring hidden and predictive information from large databases which may be business or market related. It can also be described as the process of searching for valuable information in large volumes of data. Data mining is relatively a powerful new technology with great potential to assist companies focus on the most important information in their data warehouses.

Data mining is a cooperative effort of humans and computers; human actually designs the databases, describe the problems and set goals while computers sort through the data and search for patterns that matches the goals.

Data Mining and Knowledge Discovery in Databases (KDD)

The idea of searching for useful patterns in data has a variety of names such as *data mining*, *knowledge extraction*, *information discovery*, *information harvesting*, *data archeology* and *data pattern processing*. The term data mining as earlier explained in section 3.0 is mostly employed by data analysts MIS specialties, statisticians and database administrators, while Knowledge Discovery in Databases (KDD) refers to the overall process of discovering useful knowledge from data; although, data mining and knowledge discovery in databases (KDD) are frequently treated as synonyms.

The term KDD was first coined by Gregory Piatetsky-Shapiro in 1989 to describe the process of searching for interesting, interpreted, useful and novel data. Reflecting the

conceptualization of data mining, it is considered by researchers to be a particular step in a larger process of Knowledge Discovery in Databases (KDD)

The knowledge discovery in databases process comprises of a few steps in chronological order that starts from raw data collections to some forms of new knowledge. This include data cleaning, data integration, data selection, data transformation, data mining, pattern evaluation and knowledge presentation.

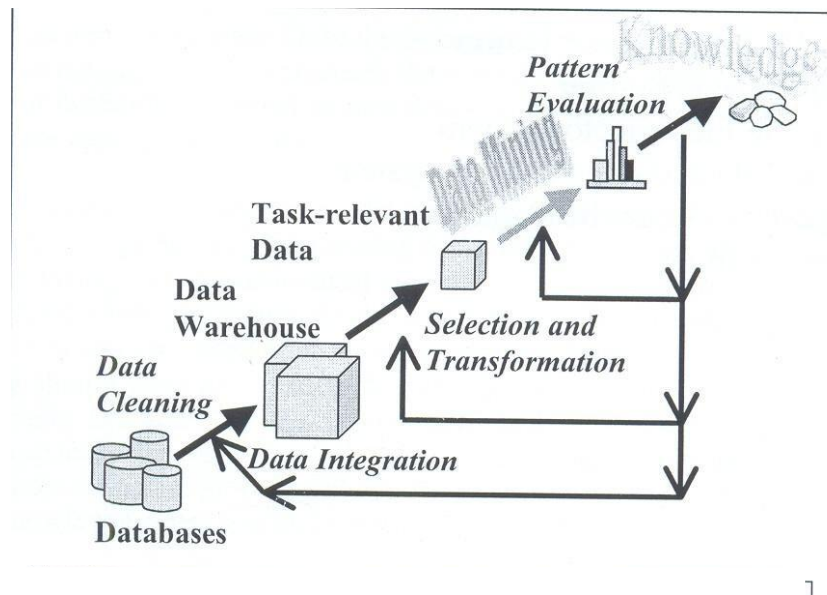


Figure 1.1 Data Mining as One of Knowledge Discovery Process
Source: *Principles of Knowledge Discovery in Databases* by O. R. Zaiane page 3

KDD being an iterative process consists of the following steps:

- ❖ **Data Cleaning:** This is also referred to as data cleansing and is a phase in which noise data and irrelevant data are removed from the collection.
- ❖ **Data Integration:** In this phase, multiple data sources which are often heterogeneous may be combined to a common source.
- ❖ **Data Selection:** The data that is relevant to the analysis is decided upon and retrieved from the data collection at this stage.
- ❖ **Data Transformation:** This is also referred to as data consolidation and is a stage where selected data are transformed into forms that are appropriate for the data mining procedure.

- ❖ **Data Mining:** This is an important step in knowledge Discovery in Databases in which clever techniques are applied for the extraction of patterns that are potentially useful.
- ❖ **Pattern Evaluation:** At this stage, patterns that are very interesting and represent knowledge are identified based on given measures.
- ❖ **Knowledge Representation:** This is the final stage of the KDD process in which the discovered knowledge is visually represented to the user. Visualization techniques are used to assist the users to have a better understanding and interpret the data mining results.

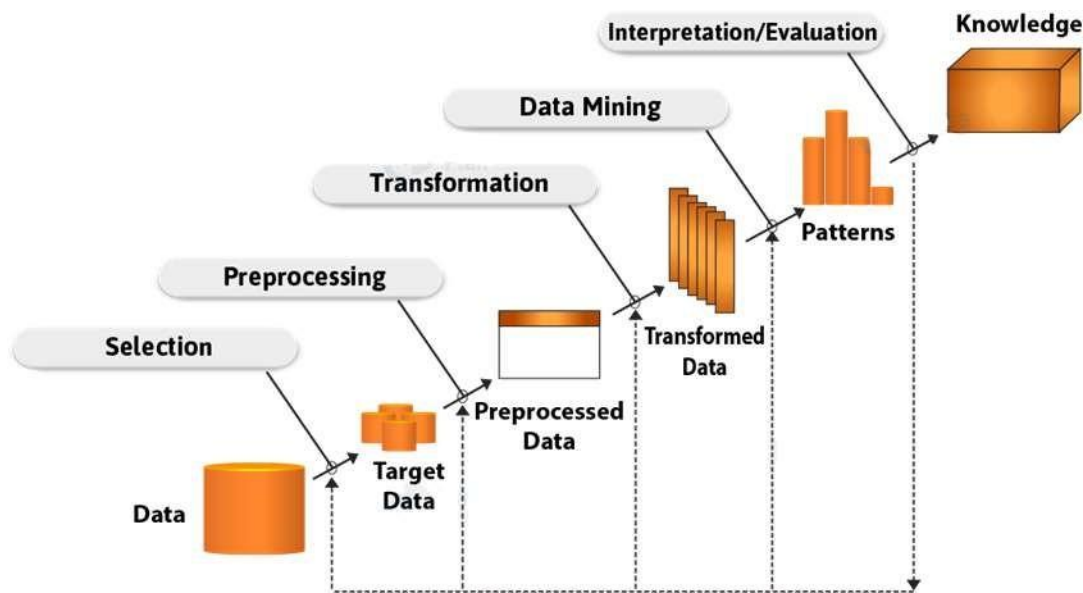
It is common to combine some of these steps together for instance, data cleaning and data integration can be performed together as a pre-processing phase to generate a data warehouse. Also, data selection and data transformation can be combined where the consolidation of the data is the result of the selection, or as for the case of data warehouses, the selection is done on transformed data.

The KDD is an iterative process and can contain loops between any two steps. Once knowledge is discovered it is presented to the user, the evaluation measures are enhanced and the mining can be further refined, new data can be selected or further transformed, or new data sources can be integrated, in order to get different and more appropriate results'

KNOWLEDGE DISCOVERY IN DATABASE (KDD)

The term *Knowledge Discovery in Databases*, or KDD for short, refers to the broad process of finding knowledge in data, and emphasizes the "high-level" application of particular data mining methods. It is of interest to researchers in machine learning, pattern recognition, databases, statistics, artificial intelligence, knowledge acquisition for expert systems, and data visualization. The unifying goal of the KDD process is to extract knowledge from data in the context of large databases.

Data mining is the core part of the knowledge discovery process. In this, process may consist of the following steps Data selection, Data cleaning, Data transformation, pattern searching (data mining), finding presentation, finding interpretation and finding evaluation. The data mining and KDD often used interchangeably because Data mining is the key part of KDD process.



KDD Process Steps

1. **Data Selection:** In this step, data relevant to the analysis task are retrieved from the database. Data can originate from multiple sources or from data warehouse. selected data must be appropriate for mining tasks
2. **Data Pre-processing:** In this step, data is transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations.
3. **Data Mining:** Intelligent operations such as Clustering, Classification, and regression are applied in order to extract data patterns.
4. **Pattern Evaluation:** Obtained results are evaluated for their accuracy. In this step, data patterns are evaluated.
5. **Knowledge presentation:** Identified patterns are presented in visual forms such as aesthetic graphs

Data Mining and OLAP

The difference between data mining and On-Line Analytical Processing (OLAP) is a very common question among data processing professionals. As we all see, the two are different tools that can complement each other.

OLAP is part of a spectrum of decision support tools. Unlike traditional query and report tools that describe what is in a database, OLAP goes further to answer why certain things are true. The user forms a hypothesis about a relationship and verifies it with a series of queries against the data. For example, an analyst may want to determine the factors that lead to loan defaults. He or she might initially hypothesis that people with low incomes are bad credit risks and analyze the database with OLAP to verify or disprove assumption. If that hypothesis were not borne out by the data, the analyst might then look at high debt as the determinant of risk. If the data does not support this guess either, he or she might then try debt and income together as the best prediction of bad credit risks (Two Crows Corporation, 2005)

In other words, OLAP is used to generate a series of hypothetical patterns and relationships, uses queries against database to verify them or disprove them. OLAP analysis is basically a deductive process. But when the number of variable to be analyzed becomes voluminous it

becomes much more difficult and time-consuming to find a good hypothesis, analyze the database with OLAP to verify or disprove it.

Data mining is different from OLAP; unlike OLAP that verifies hypothetical patterns, it uses the data itself to uncover such patterns and is basically an inductive process. For instance, suppose an analyst wants to identify the risk factors for loan default is to use a data mining tool. The data mining tool may discover people with high debt and low incomes are bad credit risks, it may go further to discover a pattern that the analyst does not consider that age is also a determinant of risk.

Although data mining and OLAP complement each other in the sense that before acting on the pattern, the analyst needs to know what would be the financial implications using the discovered pattern to govern who gets credit. OLAP tool allows the analyst to answer these kinds of questions. It is also complimentary in the early stages of the knowledge discovery process.

The Evolution of Data Mining

Data mining techniques are the results of a long process of research and product development. The evolution started when business data was first stored on computers with data access improvements and generated technologies that allow users to navigate through their data in real time. This evolutionary process is taken beyond retrospective data access and navigation to prospective and proactive information delivery.

Data mining is a natural development of the increased use of computerized databases to store data and provide answers to business analysis. Traditional query and report tools have been used to describe and extract what is in a database. Data mining is ready for application in the business community because it is supported by these technologies that are now sufficiently matured:

- ❖ Massive data collection
- ❖ Powerful multiprocessor computer

Presently commercial databases are growing at an unprecedented rate. In some organizations, such as retail, these numbers can be much larger. The accompanying need for improved computational engines can now be met in a cost- effective with parallel multiprocessor computer technology. Data mining algorithms embody techniques that have been existing for at least ten years, but have recently been implemented as nature, reliable, understandable tools that consistently outperform older statistical methods.

In the evolution data mining from business data to business information, each new step has built upon the previous one. For example the four steps listed in table 1.1 were revolutionary because they allowed new business questions to be answered accurately and quickly.

Table 1.1: Steps in the Evolution of Data Mining

Evolutionary Step	Business Question	Enabling Technologies	Product Providers	Characteristics
Data Collection (1960s)	What was my total revenue in the last five years?	Computers, tapes, disk	IBM, CDC	Retrospective, static data delivery

Data Access (1980s)	What were unit sales in New England last March?	Relational databases (RDBMS), Structured Query Language (SQL), ODBC	Oracle, Sybase, Informix, IBM, Microsoft	Retrospective, dynamic data delivery at record level
Data	What were	On line analytic	Pilot,	Retrospective,
Warehousing & Decision Support	unit sales in New England last March? Drill down to Boston.	processing (OLAP), multidimensional databases, data warehouses	Comshare, Arbor, Cognos, Microstrategy	dynamic data delivery at multiple levels
Data Mining (Emerging Today)	What s likely to happen to Boston unit sales next month? Why	Advanced algorithms, multiprocessor computers, massive databases	Pilot, Lockheed, IBM,SGL, numerous startups (nascent industry)	Prospective, proactive information delivery

(Source: *An Introduction to Data Mining*, Retrieved From: <http://www.thearling.com/text/dmwhite/dmwhite.htm>)

The core components of data mining technology have been under development for decades in research areas such as statistic, artificial intelligence and machine learning Today the maturity of these techniques coupled with high performance relational database engines and broad data integration efforts, make these technologies practical for current data warehouse environments.

Data mining derived its name from the similarities between searching for valuable business information in a large database. For example, to search for linked products in gigabytes of stored scanner data and mining a mountain for a vein of valuable ore; the two processes require either sifting through an immense amount of material or intelligently probing it to find exactly where the value resides.

If the database is given a sufficient size and quality, data mining technology can generate new business opportunities by providing the following capabilities:

- ❖ **Automated Prediction of trends and Behaviors**: Data mining automates the process of searching for predictive information in large databases. Questions that may traditionally require extensive hands-on analysis can now be answered directly from data very quickly. An example of a predictive problem is targeted marketing. Data mining uses data on past promotional mailings to identify the most likely target to maximize return on investment in future mailings. Other predictive problems include forecasting bankruptcy and other forms of default, and identifying segments of a population likely to respond similarly to given events.
- ❖ **Automated Discovery of Previously Unknown Patterns** : Data mining tools sweep through databases and identify previously hidden patterns in one step. An example of pattern discovery is the analysis of retail sales data to identify seemingly unrelated products that are often purchased together. Other pattern discovery problems include detecting fraudulent credit card transactions and identifying anomalous data that could represent data entry keying errors.

Data mining techniques can yield the benefits of automation on existing software and hardware platforms and can be implemented on new systems as existing platforms are upgraded and new products developed. When data mining tools are implemented on high performance parallel processing systems, they can analyze massive database in minutes. Faster processing means that users can automatically experiment with more models to understand complex data. High speed makes it practical for users to analyze huge quantities of data. Larger databases in turn yield improved predictions.

Architecture for Data Mining

In order to best apply this mining technique, it must be fully integrated with a data warehouse as well as flexible interactive business analysis tools. Most data mining tools presently operate outside of the warehouse, requiring extra steps for extracting, importing and analyzing data. Moreover, when new insights require operational implementation, integration with the warehouse simplifies the application of results from data mining. The resulting analytic data warehouse can be applied to improve business processes throughout the organization, in areas such as promotional campaign management, fraud detection, and new product rollout and so on. Figure 1.2 shows architecture for advanced analysis in a large data warehouse.

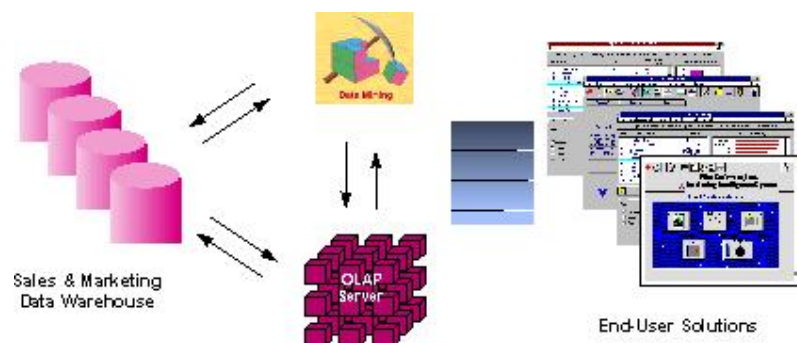


Figure 1.2 Integrated Data Mining Architecture
Source: *An introduction to Data Mining* page 5

The ideal starting point is a data warehouse that contains a combination of internal data tracking all customers contact coupled with external market about competitor's activity. The background information on potential customers also provides an excellent basis for prospecting. The warehouse can be implemented in a variety of relational database systems: Sybase, Oracle, Redbrick and so on and should be optimized for flexible and fast data access.

An OLAP (On-Line Analytical Processing) server enables a more sophisticated end-user business model to be applied when navigating the data warehouse. The multidimensional structures allow the user to analyze the data as they want to view their business- summarizing by product line, region, and other perspectives of their business. The data mining server must be integrated with the data warehouse and the OLAP server to embed ROI-focused business analysis directing into this infrastructure. An advanced, process-centric metadata template defines the data mining objectives for specific business issues like campaign management, prospecting and promotion optimization. Integration with data warehouse enables operational decisions to be directly implemented and tracked. As the warehouse continues to grow with

new decisions and results, the organization can continually mine the best practices and apply them to future decisions.

This design represents a fundamental shift from conventional decision support system. Rather than simply delivering data to the end user through query and reporting software, the Advanced Analysis Server applies users' business models directly to the warehouse and returns a proactive analysis of the most relevant information. These results enhance the metadata in the OLAP server by providing a dynamic metadata layer that represents a distilled view of a data. Other analysis tools can then be applied to plan future actions and confirm the impact of those plans (An Introduction to Data Mining)

How Data Mining Works

How does data mining tell us important things that we do not know or what is going to happen next? The technique used in performing these feats is called **modeling**. *Modeling can simply be defined as an act of building a model based on data from situations where you*

know the answer and then applying it to another situation where the answer is not known. The very act of model building has been around for centuries even before the advent of computers or data mining technology. What happens in computers does not differ much from the way people build models. Computers are loaded with lots of information about different situations where answer is known and then the data mining software on the computer must run through that data and distill the characteristic of the data that should go into the model.

And once the model is built it can be applied to similar situations where you do not know the answer.

For example, as the marketing director of a telecommunication company you have access to a lot of information such as age, sex, credit history, income, zip code, occupation and so on of all your customers; but difficult to discern the common characteristics of his best customers because there are so many variables. From the existing database of customers that contains their information as earlier mentioned; data mining tools such as neural networks can be used to identify the characteristics of those customers that make a lot of long distance calls. This then becomes the director's model for high-value customers, and he would budget his marketing efforts accordingly